



PREDICTING SONG POPULARITY FROM SPOTIFY AUDIO FEATURES: A TEMPORAL ANALYSIS OF CHANGING POPULARITY



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Abstract

This dissertation investigates to what extent Spotify audio features can be used to predict song popularity and whether the meaning of these features changes over time. The main research problem of this work is based on the question of what musical features drive song popularity. Two research questions were developed from that problem: how the structural determinants of song popularity evolved over time, and how accurately popularity can be predicted with the use of Spotify audio features.

Firstly, Exploratory Data Analysis was conducted to investigate distributions, correlations and relationships between popularity and individual audio features. Then, a cultural and temporal analysis was added, where genre and time period were treated as contextual proxies. The modelling stage started with Linear Regression as a baseline model and XGBoost Regression as a more advanced non-linear model. The Linear Regression model achieved $R^2 = 0.065$, whereas XGBoost improved the result to $R^2 = 0.121$. Hyperparameter optimisation did not produce a meaningful improvement.

A contextual XGBoost model was then created by adding genre and releaseYear to the audio features. This model achieved $R^2 = 0.342$, showing a clear improvement over the audio-only models. Temporal models were also created for pre-2000, 2000s, 2010s and 2020s. Their permutation importance results showed that the importance of audio features changed between periods. Acousticness and valence were more important in older periods, whereas loudness became the most important feature in the 2010s and 2020s.

Overall, the results show that Spotify audio features have meaningful predictive value and provide a useful foundation for modelling song popularity. Although further research and additional contextual factors are needed to improve predictive accuracy. The results also suggest that broader context, especially genre and release year, can further improve predictive performance, highlighting the value of adding contextual information alongside audio features. Finally, the temporal analysis provides evidence that the relationship between audio characteristics and popularity varies over time, showing that temporal context is an important consideration in understanding and predicting popularity.

Introduction

Background and Problem Definition

Music popularity is a phenomenon which is affected by many diverse factors. The success of a given song depends not only on its musical characteristics, but also on many external factors, such as artist popularity, promotion, being on popular playlists, social media trends, or the way streaming platforms work in general. Therefore, popularity cannot be understood solely as a consequence of the musical structure of a song. Instead, it emerges from a combination of musical, contextual and temporal influences.

Understanding these influences is important in the context listening to digital music. Streaming platforms have changed the way music is distributed and consumed, whilst providing access to large quantities of measurable information about individual tracks. Popularity within this environment can be considered a complex outcome in which the characteristics of the music interact with wider factors such as genre, release period and the context in which listeners encounter a song. Identifying patterns within these relationships can contribute to a better understanding of why some tracks achieve greater popularity than others.

At the same time, modern streaming platforms, such as Spotify, allow analysing music with the use of quantitative data. The availability of such information creates an opportunity to investigate music popularity from a data-driven perspective. This dissertation aims to investigate whether a data-driven approach can be used to identify relationships between measurable characteristics of songs and their popularity, while also considering the wider contextual factors available within the dataset.

Spotify represents tracks through measurable audio characteristics describing different aspects of their sound. These characteristics provide a quantitative representation of music that can be examined in relation to popularity. However, the availability of such measurements does not imply that popularity can be explained by audio characteristics alone.

The problem addressed by this project is not whether individual audio characteristics are associated with popular songs, but whether combining these characteristics with contextual information can provide a more complete understanding of popularity. In particular, genre and release year provide additional context that may help account for differences that cannot be captured through audio characteristics alone. This allows the project to investigate popularity as the outcome of several interacting factors rather than as a direct consequence of a song's musical properties.

Another important factor of the problem is time. Popular music is not persistent: dominant genres, production techniques, or characteristics change. Because of that, dependencies between audio features and popularity might also differ between time periods. A characteristic associated with popularity during one period may therefore be less relevant during another. Examining these temporal differences is important because it allows the analysis to consider whether relationships between musical characteristics and popularity remain consistent or evolve alongside broader changes in music.

An important limitation is the fact that popularity is a variable connected strictly to Spotify and should not automatically be treated as a universal measure of the historical, artistic or cultural significance of a song. Spotify popularity represents popularity within a particular platform and data environment rather than a complete measure of musical success. Consequently, the findings of this project should be interpreted within this context.

Research Problem and Relevance

The problem analysed in this work is important, mostly because the popularity of a song has an artistic, commercial and analytical nature. From the music industry point of view, knowledge about characteristics connected to popularity might be important for artists, producers, labels, or streaming platforms. It might support decisions of producing music, promoting, segmenting the audience, or analysing trends. Although, it does not mean that popularity can be simplified to a simple set of rules. Success of a song is also dependent on variables, that the analysed dataset does not contain, such as artist popularity, promotional budget, playlist positioning, or social media activity.

From the Data Analytics point of view, this problem is important, because it allows to implement a full analytical pipeline for cultural data. The project involves data cleaning, feature engineering, Exploratory Data Analysis (EDA), data visualisation, building regression models, judging them on their evaluation metrics, and the analysis of feature importance. Due to these factors, this paper is not limited to predicting the popularity value only but also connects the predictive approach with the result interpretation.

One thing is especially important – differencing between simply predicting popularity and understanding which features are useful for a model. The initial EDA showed that the majority of audio features do not show strong, linear relationships with popularity. Instrumentalness and acousticness were the only ones that stood out significantly – their relationship with the target variable was strong and negative. At the same time, there were strong correlations between two predictors themselves, especially energy and loudness. It suggested that popularity is a multidimensional phenomenon and needs a more complex model, rather than a simple correlation analysis.

The importance of the project is also increased by time and context. The cultural analysis showed that the genre structure of the dataset changes between time periods, and singular audio features do not stay persistent. The older periods are dominated by rock, whereas the newer ones are characterised by a larger genre variety. At the same time, one can notice that valence and track length decreased over time and instrumentalness, speechiness and danceability increased in more recent periods.

For the following reasons, the research problem is not only whether you can predict popularity or not, but also whether audio features stay stable over time. Such an approach allows to combine Machine Learning with the analysis of structural changes in contemporary music and builds a foundation for answering the research questions of the project.

Research Aim and Research Questions

The main goal of this research is to find out to what extent audio features available from Spotify can be used in explaining and predicting song popularity and whether the meaning of those features changes between different periods of time. The project then combines two approaches: predictive modelling and temporal interpretation. The first one concentrates on evaluating the accuracy of models predicting popularity, whereas the second one on the analysis of changes in the meaning of respective audio features.

Two research questions have been formulated:

RQ1: How have the structural determinants of song popularity evolved over time in contemporary music?

RQ2: How accurately can song popularity be predicted using Spotify audio features?

The first research question involves changes in time. It assumes, that features important for predicting popularity may not stay stable, because music genres, production techniques and the general structure of music changes over time. It was researched in the further parts of the project by building separate models for different periods and comparing their feature importance.

The second research question concentrates on the prediction accuracy. A linear Regression model was used as a basis, and an XGBoost model was used as a more advanced non-linear model. Then, an additional model that uses more variables (genre, releaseYear) was developed to check whether additional context improves the predictive performance.

In this way, the two research questions are complementary: one focuses on how accurately popularity can be predicted, while the other examines whether the influence of the predictor's changes over time.

Research Approach

To answer both research questions, this project uses a multi-step Data Analytics pipeline. The process started with preparing and cleaning the data. Two initial .csv files has been joined, and then missing values, duplicate rows, duplicate track IDs, datatypes, and release date correctness were all checked. Next, new time features were introduced, including releaseYear, releaseMonth, and timePeriod. Also, redundant columns were removed to simplify the structure of the dataset.

Subsequently, Exploratory Data Analysis was conducted, which contained the analysis of variations, correlations and relations between popularity and audio features. EDA was also used to identify the limitations of the dataset: it contained way more songs from 2010s and 2020s compared to the rest of the periods, as well as weak relationships between the majority of individual audio features with popularity.

After the interim submission, the pipeline has been expanded to include cultural and temporal analysis. In this part the following were analysed: changes in the genre variable structure, mean genre popularity, and audio characteristics in different time periods. Genre and time period were treated as contextual proxies, which allowed to include a broader musical context without stating that the dataset directly measures the culture,

The predictive modelling stage started with building a Linear Regression model as a baseline. Next, an XGBoost Regression model was built to see whether a non-linear model shows the relationships between popularity and audio features in a better way. Additionally, a hyperparameter optimisation was utilised with the use of RandomizedSearchCV. The next step was expanding XGBoost with genre and releaseYear variables, which allowed to compare the previous model that was only based on audio features with a model that included additional context.

The last stage was temporal modelling. The dataset was divided into four broader periods: pre-2000, 2000s, 2010s and 2020s. For each period, a separate XGBoost model was trained with the use of the same audio features. Next, the importance of each audio feature was calculated using XGBoost's built-in feature importance scores. These scores were then compared across the four time periods to examine whether the relative influence of individual audio features on popularity changed over time. That allowed to compare the meaning of respective predictors between time periods and directly address the first research question.

The entire pipeline combines three elements: data preparation, prediction, and interpretation, which allows to not only compare the accuracy of the models, but also investigate changes in time.

Structure of the Dissertation

This dissertation is organised into six chapters, each corresponding to a particular stage of the research. Following this introductory chapter, Chapter 2, Literature Review, presents earlier research on predicting song popularity, the usage of audio features, temporal changes, and the role of broader genre and cultural context.

Subsequently, Chapter 3, Data and Methods, describes the source of the data, structure of the dataset, process of data cleaning, feature engineering, EDA, cultural and temporal analysis and building the models. In this chapter, evaluation metrics, hyperparameter optimisation, and permutation importance are also described.

Chapter 4, Results, presents the most important EDA, cultural analysis and predictive modelling results. It includes the comparison of Linear Regression, XGBoost, the contextual model and finally the models built for respective time periods.

In Chapter 5, Discussion, the results are interpreted in the context of the research questions and previous literature. There is an emphasis on the limited predictive power of audio features themselves, the improved performance after adding genre and release-year variables, and changes in the meaning of certain audio features over time.

Chapter 6, Conclusion and Future Work, summarises the most important results, answers both research questions and shows the biggest project limitations. It also prompts possible further research on the topic.

Literature Review

Music Popularity as a Data Analytics Problem

The popularity of music is a tough analytical problem, because there is no such thing as one characteristic that determines whether a song will be successful, or not. Popularity is based on factors such as characteristics of the song, but also on external factors, such as artist popularity, promotion, playlist presence, or streaming platform environments. From the Data Analytics perspective it means that it is a problem in which many factors can influence the target variable value.

The development of streaming platforms allowed to describe music in a form of quantitative data. Audio features present the characteristics of a song, such as energy, danceability, loudness, valence, acousticness, or instrumentality in a form that is appropriate to use in statistical analysis and Machine Learning. Due to these factors, one can investigate not only relations between these features and popularity, but also how important they are in predictive modelling.

Interiano et al. (2018) show, that the presence of measurable characteristics does not mean, that musical success can be easily predicted. The authors stated that they were not able to exceed a prediction accuracy of 0.74. The result shows, that audio characteristics have predictive value, but are not the full explanation for popularity.

Saraigh (2023) came to a similar conclusion and states that both of his Machine Learning models yield a predictive power of 69%. It shows that Machine Learning might be efficiently used in music popularity prediction, but its results should be interpreted as a partial explanation of a more complex phenomenon. Such an approach forms an analytical foundation for the rest of this research.

Audio Features and Popularity Prediction

One of the main research directions on music popularity is checking if measurable audio features can be used for predicting the success of a song. In this approach variables such as loudness, energy, danceability, acousticness, instrumentality, or valence are treated as predictors, where popularity or music chart presence are target variables.

Interiano et al. (2018) analysed over 500,000 songs from Great Britain released between 1985 and 2015, comparing the songs that appeared on charts with the ones that did not make it. The study showed that successful songs were different from the rest in terms of the musical characteristics, but the audio characteristics alone were not enough to make very accurate prediction. After adding information about previous artist popularity, the accuracy rose from 0.74 to 0.86, which shows the meaning of external factors.

Saragih (2023) analysed over 90,000 observations connected to Spotify users in Indonesia. The best regression model achieved $R^2 = 0.6857$ and $RMSE = 12.33$. Among the most important features were release date, loudness, instrumentality, acousticness, duration, speechiness and energy.

Zangerle et al. (2019) also showed that prediction may be more accurate if combined with different types of information. Their model utilised both low-level and high-level audio features. They stated that “the release year information is the most important high-level feature” (p. 324), which furthermore emphasises on the temporal context in predicting popularity.

Overall, the results of these studies suggest that audio features have a real predictive value, although their accuracy rises after including additional context information.

Temporal Changes in Music

Another important element of the study of music popularity is changes in time. Features related to the success of a song may not be constant, because as the time passes, production techniques, dominant genres and audience preferences change. Because of that, analysing one global model might not be sufficient to fully understand the relationship between audio features and popularity.

Interiano et al. (2018) showed that the characteristics of popular songs changed over analysed time. In the study they noticed changes in features such as happiness, brightness, or sadness, which suggests that the characteristics are not stable over time.

The temporal context is also important on the genre level. Petitbon and Hitchcock (2022) analysed the changes in genre popularity between 1974 and 2018. One of the observed trends was the decline of rock popularity in the late 1980s through 2000s. Such inclines or declines in popularity might also affect the characteristics present in popular music.

The significance of time also appears in Zangerle et al.'s (2019) work, where release year was specified to be an important high-level feature. Overall, the results of these studies support the approach in which popularity prediction should be supplemented with temporal analysis, and feature importance should be compared between different time periods.

Genre and Cultural Context

The popularity of music is not dependent on characteristics of one song only, because music functions in a broader genre and cultural context. Musical genres might represent different styles of production and sound structure, and their popularity might change with the preferences of listeners. Because of that, genre might deliver an additional context for the popularity analysis, which audio features alone do not represent.

Petitbon and Hitchcock (2022) analysed the genre popularity, utilising data from *Billboard Hot 100* and *Village Voice Pazz & Jop*. The authors emphasised on a broader character of their approach and stated that their analysis focused on cultural, societal, and demographic factors influencing genre preference across a specific culture. Their results show that the role of respective genres changes over time, as can be seen in the fall of popularity of rock and a rise of popularity of hip-hop.

At the same time, genre should not be treated as a direct and objective measure of culture. Petitbon and Hitchcock (2022) note that the genre classification is imperfect, because songs can belong to more than one category, and their classification is dependent on the method of the interpreter. In the context of this research, genre is then treated as contextual proxy, which can help in analysing musical changes, but does not measure the culture directly. Such an approach is important in the later interpretation of cultural and temporal analysis.

Limitations of Existing Approaches

So far the studies show that predicting music popularity with using audio features has some significant limitations. The most important one is the fact that there is no information on external factors, such as previous artist popularity, promotion, playlist presence, or social media activity.

Another limitation is the way popularity itself is defined. In different studies the target variable might mean the music chart presence, belonging to a certain popularity class, or a quantitative Spotify popularity score. It means that the results of different models not always can be directly compared.

The context of the data used is also important. Saragih (2023) analysed the data of Indonesian Spotify users, and because of that his results might reflect that specific market. On the other hand, other studies show Billboard or nation-wide musical charts. Consequently, predictive performance and feature importance might heavily rely on the method used, but also on the source of the data, the timeline, and the definition of musical success.

These limitations suggest that the model results should always be interpreted carefully and always in the context of the dataset on which they were built.

Research Gap

The literature review shows that previous studies on music popularity might be divided into a few different directions. Some of the work concentrates mostly on predictive modelling and checking whether the audio features can predict popularity and/or be on music charts. Other work concentrates more on changes that happen over time, such as evolution of musical genres, audience preferences, or characteristics of popular music. There are also approaches that combine audio features with additional high-level variables, such as genre, release year or artist popularity. However, these directions are often considered separately. Predictive studies mainly focus on achieving satisfactory model performance and identifying which variables contribute to prediction, without examining whether these relationships remain consistent across different periods of music. In contrast, studies concerned with temporal changes tend to describe how genres, listening preferences or musical characteristics evolve over time, rather than testing how such changes affect the ability to predict popularity. Similarly, studies incorporating contextual variables demonstrate that information beyond audio characteristics can be useful, but do not necessarily examine how the importance of these predictors varies between different periods.

The main research gap does not arise from the lack of music popularity studies, but from the limited number of analyses that combine prediction, temporal comparison and the interpretation of feature importance. Taken together, the existing research therefore provides useful evidence about either predictive performance or changes in music over time but offers less insight into how these two aspects interact. There is limited consideration of whether the characteristics that are useful for predicting popularity remain equally important across different periods. The evaluation of model performance alone does not show whether the characteristics remain equally relevant in different periods. On the other hand, the analysis of changing genres or musical trends does not answer the question of how accurately popularity can be predicted.

This dissertation combines both approaches. Firstly, the predictive value of Spotify audio features using Linear Regression and XGBoost is evaluated. Then, the meaning of the same features across different periods of time is compared. Additionally, genre and release year are inspected to see if the predictions are better compared to the model with audio-only features.

This way, the project is not limited to the question of whether popularity can be predicted but also analyses whether the structure of the information used in such prediction changes over time. Therefore, the dissertation connects predictive performance with the temporal stability and interpretation of the predictors, addressing both research questions within a single analytical framework.

Data and Methods

Data Source and Collection

The data used in this work comes from a publicly available Spotify Music Dataset published on the Kaggle platform. The dataset contains descriptive song information and a set of Spotify audio features, which can be used to quantitatively analyse the data. The data has been downloaded in two .csv files.

The data was neither collected directly from the Spotify users nor through web scraping or APIs. The project utilises an existing dataset shared by a third-party website, so the analysis is based only on the files included on the website. It also means that the work is limited to the scope and quality of the data from the source dataset.

Initial Dataset Description

Before starting the data cleaning, the data was stored in two separate .csv files. The first file contained 1,686 observations, whilst the second had 3,145 observations. Both of them contained the same 29 columns. The structure of the variables was the same for both files, but the order of the columns was different in the second file, so they were normalised before concatenation. After running `pd.concat()`, one set was created with 4,831 records.

The initial variables contained descriptive information, Spotify IDs, playlist information and audio features. Amongst the descriptive variables there was data such as track_name, track_artist, track_album_name, track_album_release_date and playlist_genre. The dataset also contained technical IDs, such as track_id, track_album_id and playlist_id. Audio features were also present in the dataset (*table 1*).

Feature	Description
energy	A measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy.
tempo	The speed of a track, measured in beats per minute (BPM).
danceability	A score describing how suitable a track is for dancing based on tempo, rhythm stability, beat strength and overall regularity.
loudness	The overall loudness of a track in decibels (dB). Higher values indicate louder tracks overall.
liveness	The likelihood of a track being performed live. Higher values suggest more audience presence.
valence	The overall musical positiveness (emotion) of a track. High valence sounds happy; low valence sounds sad or angry.
speechiness	Measures the presence of spoken words.
instrumentalness	The likelihood a track contains no vocals. Values closer to 1.0 suggest solely instrumental tracks.
acousticness	A confidence measure of whether a track is acoustic (1) or not (0).
mode	Indicates the modality of the track.
key	The musical key, represented as an integer from 0 to 11, mapping to standard Pitch class notation.
duration_ms	The length of the track in milliseconds.
track_popularity	A score (0–100) which is calculated based on total number of streams in relation to other songs.

Table 1 - Most Important Features

The initial inspection showed the presence of missing values, duplicate rows, duplicate track IDs and problems with the format of release dates. Therefore, data cleaning was needed before proceeding to EDA.

Data Cleaning

The process of data cleaning started with checking for missing values. Initially in the dataset there were columns/rows that contained singular missing values. Because their number was substantially smaller than the entire dataset, records with missing values were removed. Then, exact duplicates were checked. 43 exact duplicates were detected, which were also removed.

The next step was the control of duplicate track IDs. Even after removing the exact duplicates, the dataset still contained 293 repeating track IDs. To make each song being represented only once, each subsequent appearance of a song was removed, i.e., only the first appearances stayed in the dataset. After that the dataset contained 4,493 observations.

Subsequently, datatypes of some variables were changed. `time_signature`, `track_popularity`, `mode`, `key` and `duration_ms` were converted to integers. Additionally, `duration_ms` was converted to minutes by creating a new `duration_min` variable, rounded to two decimal places.

There was an emphasis on the `track_album_release_date` variable, as it was key for future temporal analysis. The dates were converted to the `datetime` variable type. As a result, 125 records with incorrect/incomplete dates were detected, which were removed from further analysis.

At the end, columns that did not contribute to the analysis as a whole were also removed from the dataset. That included URLs, technical IDs, `playlist_name`, `track_album_name` and `playlist_subgenre`. The rest of the variables were tidied up and renamed to make it easier to reference in the future analysis, e.g., `track_popularity` to `popularity`, or `playlist_genre` to `genre`. After finishing the cleaning, the dataset has been saved as a new `.csv` file and used in future stages.

Feature Engineering

After completing the data cleaning stage, additional variables were created that were used in further analysis. Based on `releaseDate`, more specific variables were isolated: `releaseYear` and `releaseMonth`. Next, based on `releaseYear`, the `timePeriod` variable was defined and divided into pre-1980, 1980s, 1990s, 2000s, 2010s and 2020s.

During the EDA stage a `popularityGroup` variable was defined to make classification and comparison of audio features simpler and clearer. Songs with popularity lower than 40 were classified as Low popularity, values between 40 and 69 were Medium popularity, and values 70 and above were named as High popularity. This variable was used only for descriptive analyses purposes and was not utilised as a predictor in regression models.

At the temporal modelling stage another variable called `modelPeriod` was created, diving the data into Pre-2000, 2000s, 2010s and 2020s. Such broader division was implemented due to the fact that initial older time period sample sizes were not sufficient for accurate analyses.

In the modelling stage, the `genre` variable was converted using one-hot encoding, so it could be used with numerical audio features in XGBoost models (*table 2*).

Source variable	Engineered feature	Description
releaseDate	releaseYear	Year of release extracted from the release date.
releaseDate	releaseMonth	Month of release extracted from the release date.
releaseYear	timePeriod	Six-category temporal grouping used for descriptive and cultural analysis.
popularity	popularityGroup	Low, Medium and High popularity categories used for descriptive analysis.
releaseYear	modelPeriod	Four broader temporal groups used for temporal modelling.
genre	Genre dummy variables	One-hot encoded genre categories used by the contextual XGBoost model.

Table 2 - Engineered Features

Exploratory Data Analysis

After preparing the dataset, Exploratory Data Analysis was conducted to understand the data structure, distribution of variables and correlations between audio features and popularity. The analysis contained descriptive statistics, distribution plots, correlation analysis and comparison between previously created popularity groups.

Firstly, the distribution of popularity, the scope of releaseYear and the number of songs in each time period were analysed. It allowed to identify an important temporal imbalance – the majority of observations came from 2010s and 2020s, whereas the older periods were represented by much smaller sample size. The genre distribution was also inspected to identify which genres are most present in the dataset.

Subsequently, the correlation matrix was calculated for popularity, releaseYear and the main audio features. That analysis allowed to identify simple linear relationships and potential correlations between the predictor variables. Additionally, scatter plots for individual audio features and popularity were created. That was to show potential clear individual correlations.

At the end, mean values of audio features for low, medium and high popularity groups were compared. EDA primarily acted as a diagnostic and preparation tool before the modelling stage to show the most important patterns and limitations in the data.

Cultural and Temporal Analysis

After finishing the initial EDA, a cultural and temporal analysis was conducted. Its goal was to extend the project in terms of the genre and temporal context. The dataset did not contain any direct cultural variables, genre and timePeriod were treated as contextual proxies and not direct measurements of culture.

The first step was the analysis of genre composition over time. To make the results more readable, the ten most occurring genres were left as separate categories, whereas the rest was joined into the “other” group. The percentage of overall contribution to the given time period was then calculated. That allowed to compare different time periods, even though the sample sizes were not equal.

The next step was the analysis of the mean popularity based on genre and timePeriod. To limit the influence of very small groups, the genre-period combinations containing less than five songs were eliminated from the interpretation.

The last part contained the comparison of audio characteristics over time. Ten audio features were standardised, which allowed to analyse their relative changes on the same scale, despite the initial scale differences in some variables. Then mean values for each time period were calculated. At the end, the oldest and the newest period were compared.

Predictive Modelling

The predictive modelling stage has been designed to answer the second research question on the accuracy of popularity predictions. Ten continuous audio features were selected for main modelling: energy, danceability, loudness, liveness, valence, speechiness, instrumentalness, acousticalness, tempo and durationMin. Key, mode and timeSignature variables were skipped in the main models.

The dataset was split using a train/test split (80/20) with the random_state parameter of 42 for reproducibility. The training set contained 3,494 observations, whereas the test set contained 874. The distribution of popularity in both parts was similar.

The first model was Linear Regression, used as a baseline. Audio features were re-scaled using StandardScaler before the training. Then, XGBoost Regression was used to see if a non-linear model is better suited to the relations between the predictors and the target variable.

For XGBoost a simple hyperparameter optimisation was also conducted using RandomizedSearchCV and a 3-fold cross-validation. Then, an additional contextual model was built in which releaseYear, and genre variables were added. Genre was previously transformed using one-hot encoding, which increased the number of predictors to 46. To make all models comparable, the same training and test observations were used throughout the modelling stage.

Model Evaluation and Feature Importance

To evaluate the predictive performance, three standard metrics for regression problems were used: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 . MAE shows the mean absolute error of the prediction, whereas RMSE penalises larger mistakes. R^2 was used to evaluate the explainable variation of the popularity in the model. The set of these metrics allowed to compare Linear Regression, base XGBoost, optimised XGBoost, and the contextual model.

To answer the first research question, different temporal models were created. Due to the small sample size in older time periods, four groups were created: pre-2000, 2000s, 2010s, and 2020s. The same set of audio features and the same XGBoost configuration were used for each period to ensure smooth result comparison.

The meaning of predictors was evaluated based on permutation importance. This method measures the change of performance of a model after randomly shuffling the value of a single feature. If after such shuffling the model performs way worse, it means that the given variable was more useful for the prediction. The importance was calculated independently for each period using `n_repeats = 10` and then the results were combined into one table and one heatmap.

Software and Ethical Considerations

The entire analysis was conducted in the Jupyter Notebook environment using the Python language. To prepare and process the data, mainly pandas and numpy libraries were used. The visualisations were created with the aid of matplotlib and seaborn. The modelling stage used scikit-learn to train/test split, StandardScaler, LinearRegression, evaluation metrics, RandomizedSearchCV, and permutation_importance. The non-linear model was built with the aid of the xgboost library and the XGBRegressor class.

From the ethical point of view, the dissertation did not involve any user contact. It also did not process any personal data. The dataset is publicly available and contains information on songs. It does not contain any information on individual Spotify users, their listening history, or anything that allows their identification.

The interpretation of the target variable (popularity) is also important. It is a measure associated with the Spotify streaming platform, so it should not be treated as a universal indicator of cultural value or historical success of a given song. The results of this work should be interpreted within the scope of the used dataset and the context of the Spotify platform.

Results

EDA Results

After finishing the data cleaning stage, the final dataset contained 4,368 songs. The mean value of popularity was ~ 53.1 , with the median of 53. The distribution of popularity was bimodal, with clear peaks around 50 and 70 (*figure 1*). The most songs fell into the medium popularity category which contained 2,093 observations. The high popularity category contained 1,169 songs, whereas the low popularity group had 1,106 of them.

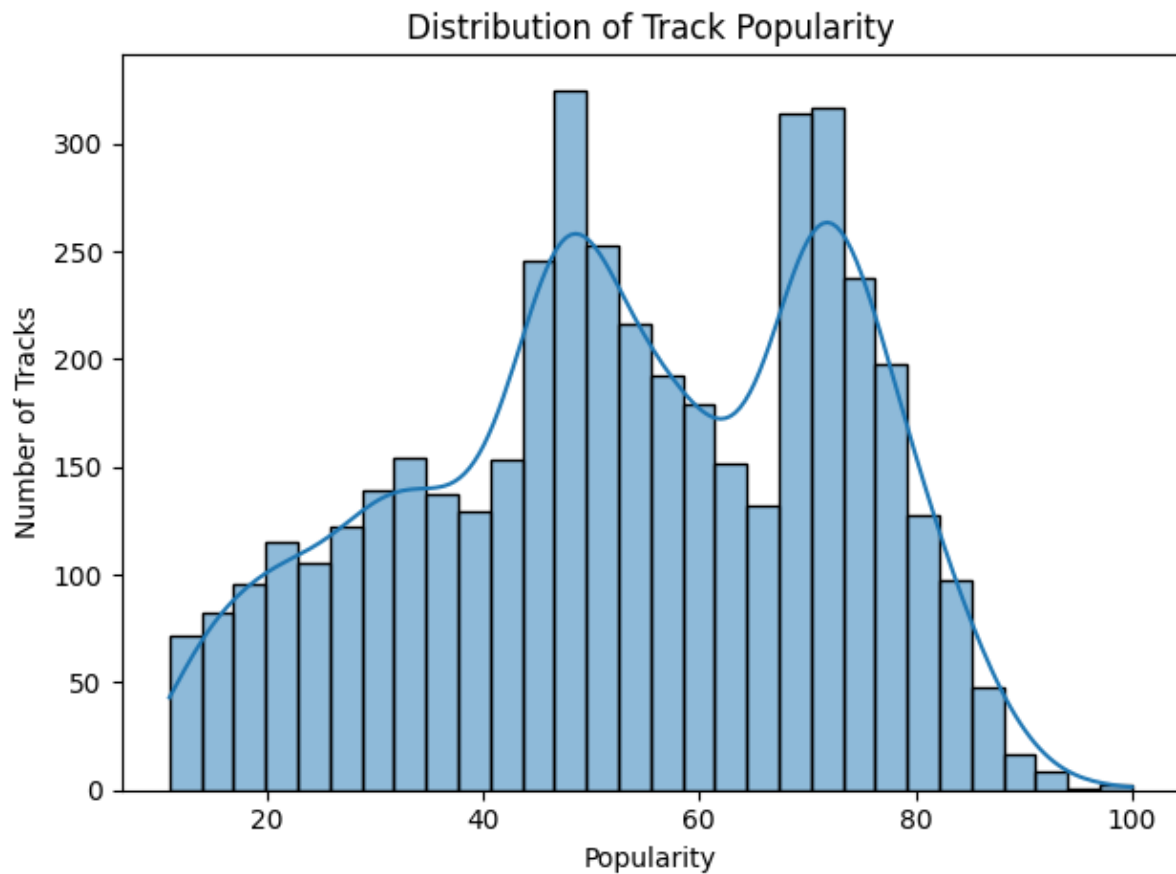


Figure 1 - Distribution of Track Popularity

EDA also showed a strong temporal imbalance. Out of all observations, 2,799 songs came from 2020s. Only 47 came from pre-1980 and 67 from 1980s (*figure 2*). It means, that the modern music is greatly overrepresented in the dataset.

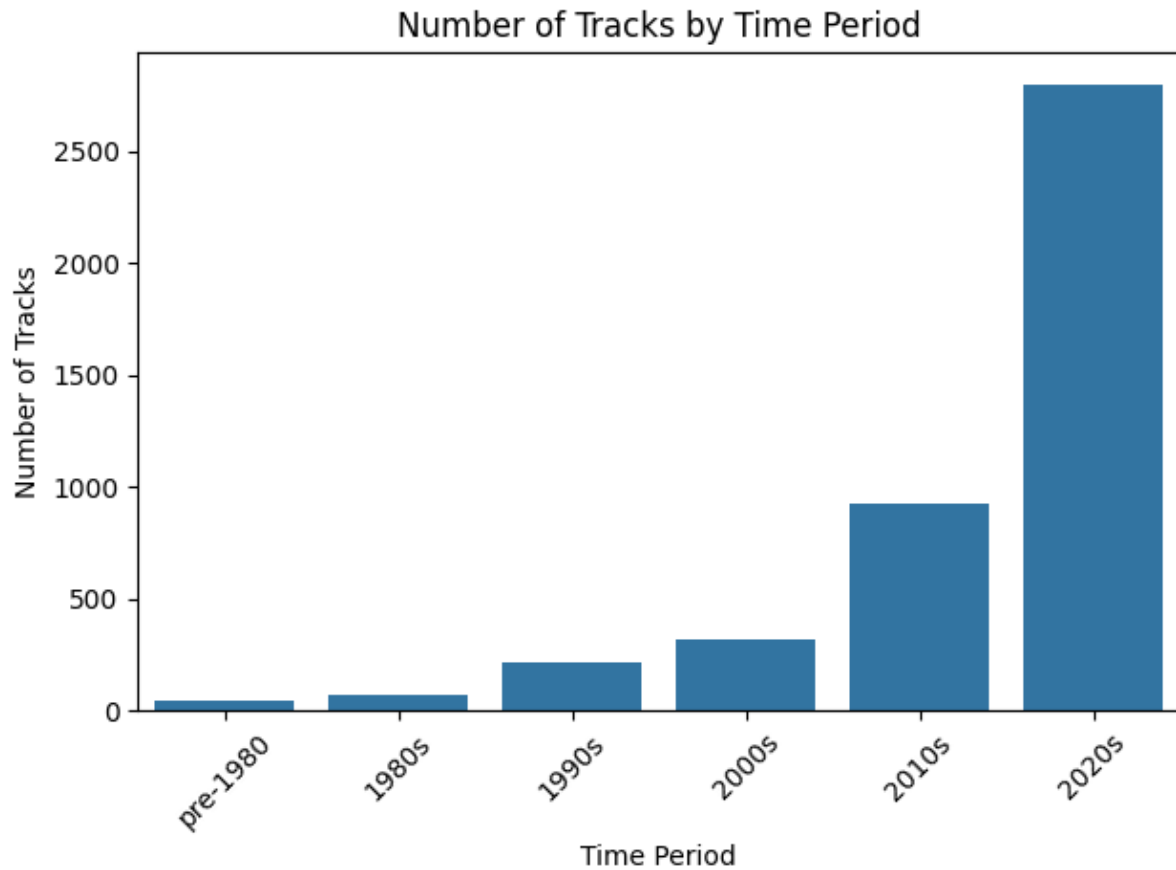


Figure 2 - Number of Track by Time Period

The correlation analysis (*figure 3*) did not show any strong linear relationships between the majority of audio features and popularity. The significant negative correlations included instrumentalness and acousticness. At the same time, the strongest relationship between the predictors themselves appeared between energy and loudness, with the correlation of ~ 0.8 .

The comparison of popularityGroup also showed that highly popular songs had lower instrumentalness and acousticness on average, where the differences between the majority of the other features were smaller.

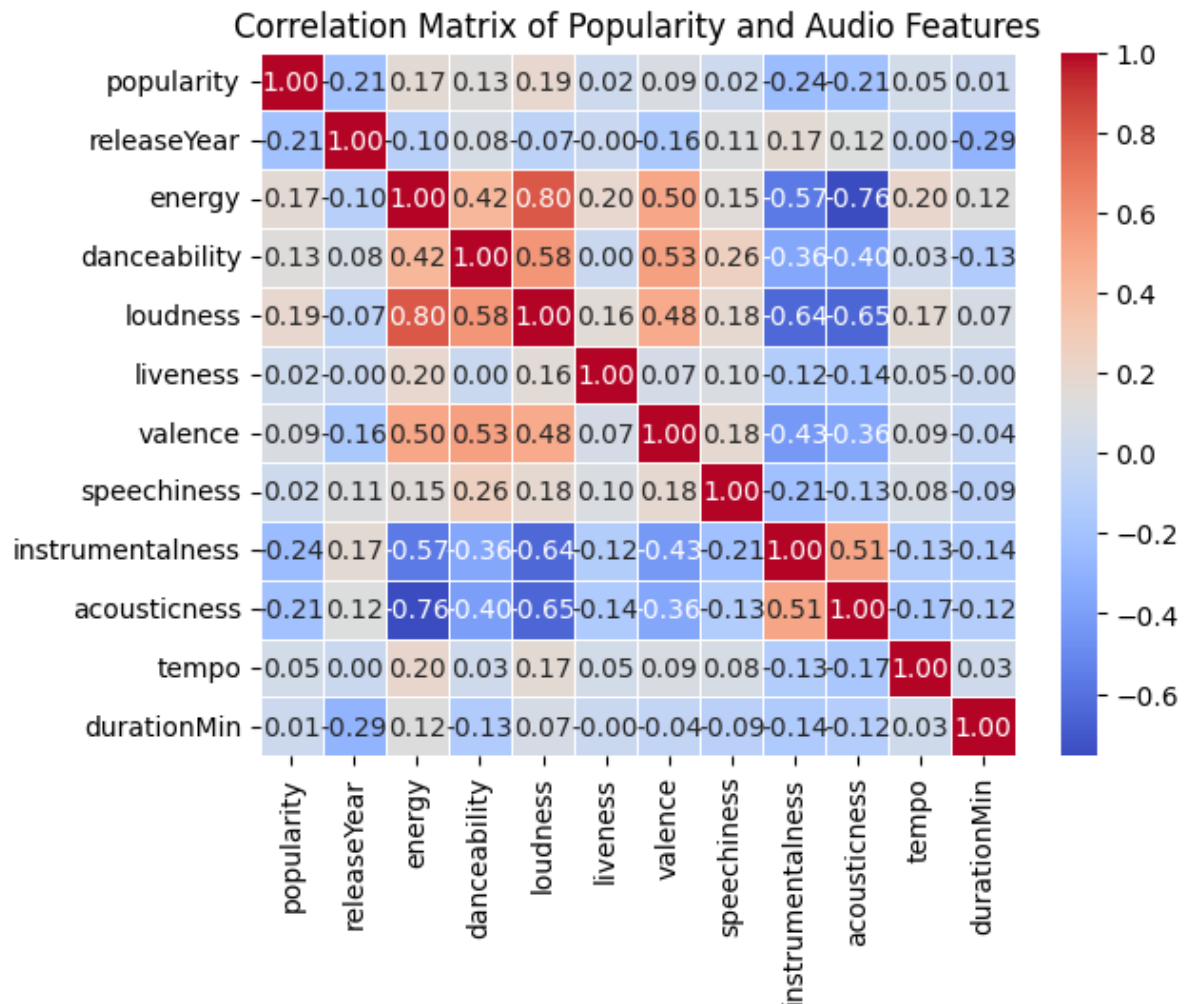


Figure 3 - Correlation Matrix of Popularity and Audio Features

Cultural and Temporal Results

The cultural and temporal analysis showed clear differences in the genre structures between time periods. Rock was dominant in the older periods. It represented ~66% of songs in pre-1980 and ~64% of songs from 1980s. In the next decades, the structure became more diverse. In 2010s, electronic and pop were the most represented. In 2020s, electronic, latin, lofi, ambient, and hip-hop were the most numerous (*figure 4*).

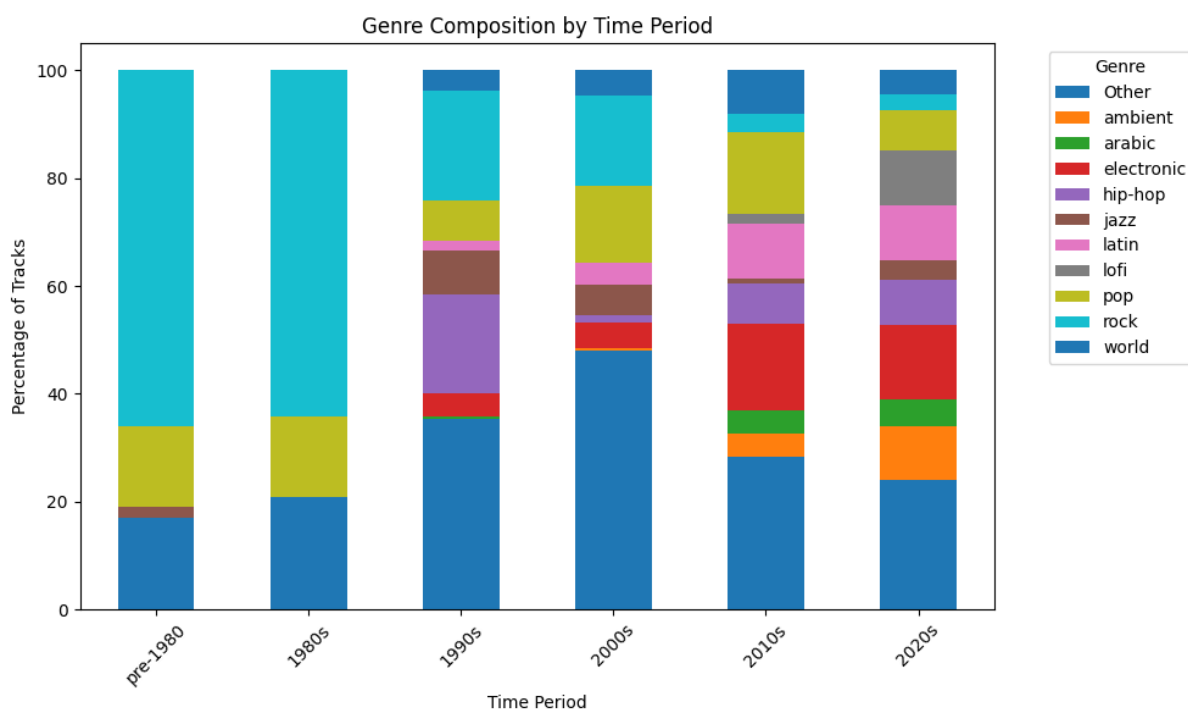


Figure 4 - Genre Composition by Time Period

The differences were also visible in the mean of popularity in respective time periods. For example, older pop and rock songs present in the dataset had higher Spotify popularity scores than their newer equivalents. This results have to be proceeded with caution though, because older time periods contain less observations and can be more prone to survivorship bias. In the genre-period analysis, combinations of less than 5 songs were excluded.

The temporal comparison of audio features also showed visible changes. Between pre-1980 and 2020s, valence and durationMin significantly dropped. On the other hand, a rise was observed in instrumentalness, speechiness and danceability. Energy changed little over time, but higher than normal values can be seen in 1980s (*figure 5*).

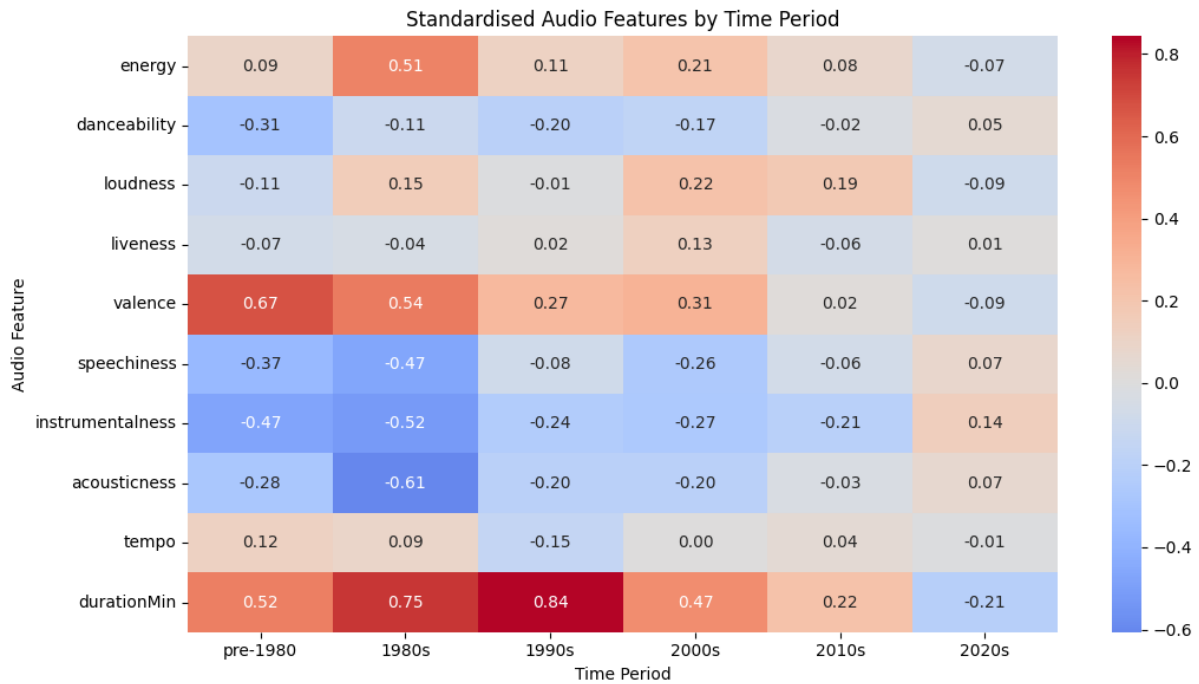


Figure 5 - Standardised Audio Features by Time Period

Baseline and XGBoost Performance

Three evaluation metrics were used to assess model performance: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). MAE represents the average absolute difference between the predicted and actual popularity scores and is expressed in popularity points, making it directly interpretable. RMSE is also expressed in popularity points but gives greater weight to larger prediction errors, making it useful for identifying models that make particularly inaccurate predictions. R^2 measures the proportion of variation in popularity explained by the model, where higher values indicate greater explanatory power. These metrics were selected together because they provide information about average prediction error, larger prediction errors, and the overall amount of variation explained by each model.

The first model was Linear Regression, which was the baseline model. It achieved the following results: MAE = 16.132, RMSE = 19.261 and $R^2 = 0.065$. The MAE indicates that predictions differed from the actual Spotify popularity score by approximately 16.1 points on average, while the higher RMSE of 19.3 indicates the presence of some larger prediction errors. The R^2 of 0.065 means that the model explained ~6.5% of the variation in popularity. The linear model therefore explained only a little part of the overall variation in popularity, and its predictions showed relatively high error given that Spotify popularity is measured on a 0-100 scale.

Then, XGBoost Regression was implemented. The model achieved the following results: MAE = 15.201, RMSE = 18.673 and $R^2 = 0.121$. Compared to the baseline model, all three metrics improved. The average prediction error decreased by ~0.93 popularity points, while RMSE decreased by ~0.59 points. More importantly, R^2 increased from 0.065 to 0.121, meaning that the XGBoost model explained approximately 12.1% of the variation in popularity compared with 6.5% for Linear Regression. It means that the non-linear model showed the relations between

the audio features and popularity more effectively than the linear baseline, although a large proportion of the variation remained unexplained.

After improving XGBoost with hyperparameter optimisation, it achieved MAE of 15.33, RMSE of 18.688 and R^2 of 0.12. Hyperparameter optimisation was performed using RandomizedSearchCV. Five parameter combinations were evaluated using three-fold cross-validation, resulting in 15 model fits, with RMSE used as the optimisation criterion. The best parameter combination used 200 estimators, a learning rate of 0.03, maximum tree depth of 3, subsample of 0.8, gamma of 0.2 and column subsampling of 0.8. The best cross-validation RMSE was 18.17.

However, when the optimised model was evaluated on the unchanged test set, it did not improve upon the original XGBoost model. MAE increased slightly from 15.201 to 15.33, RMSE increased from 18.673 to 18.688, and R^2 decreased marginally from 0.121 to 0.12. Therefore, the optimisation did not produce a meaningful improvement in generalisation performance. This suggests that the original XGBoost configuration was already performing close to the best achievable result within the tested parameter space and, more importantly, that model configuration alone was not the main factor limiting predictive performance.

The results show that XGBoost had better predictions than the baseline linear model, but its predictive power stayed limited, due to containing audio features only (*figure 6*). Nevertheless, the improvement over Linear Regression demonstrates that modelling non-linear relationships provides additional predictive value. The limited size of the improvement also provides a justification for the subsequent contextual model, in which genre and release year were added to investigate whether information beyond audio characteristics could further improve performance.

	Model	MAE	RMSE	R2
0	Linear Regression	16.132	19.261	0.065
1	XGBoost	15.201	18.673	0.121
2	Optimised XGBoost	15.330	18.688	0.120

Figure 6 - Model Comparison

Contextual Model Results

To check if additional context improves predictions, an XGBoost with the addition of releaseYear and genre was built. After implementing one-hot encoding the model contained 46 predictors in total. The same training and test observations as in the audio-only model were used.

The contextual model achieved the following results: MAE = 12.796, RMSE = 16.153 and R^2 = 0.342. Compared to audio-only XGBoost (which scored 0.12 in R^2), the increase is 0.223. MAE and RMSE decreased by ~2.53 points.

It was the largest performance improvement across all models. The results show that the information on genre and releaseYear were providing the model significant additional context. At the same time, the model still did not explain the majority of variation in popularity, so even after implementing the improvement, it stayed limited.

Temporal Model Performance

To see if the predictive performance changes between periods, different XGBoost models were built for pre-2000, 2000s, 2010s and 2020s. The results were significantly different between the groups.

The best results achieved the 2000s model with the following: MAE = 10.524, RMSE = 13.533 and $R^2 = 0.383$. The pre-2000 model achieved $R^2 = 0.209$, MAE = 11.042 and RMSE = 15.116. For more recent periods, the predictive performance was lower: the 2010s model achieved R^2 of 0.125 and the 2020s' R^2 was 0.149.

The results show that the dependency between audio features and popularity was not equally strong in all periods. The performance of the 2000s' model was especially better in comparison to 2010s and 2020s model.

One has to remember about the sample size, though. Pre-2000 contained 326 songs, 2000s 317, 2010s 926, and 2020s had them 2,799. Because of that, a direct comparison demands caution.

Feature Importance Across Time

The permutation importance analysis showed that the meaning of respective audio features changed between studied periods. For pre-2000, the highest importance had acousticness (1.775), and then valence (0.667) and liveness (0.398). In 2000s, acousticness was also dominant (3.919). Valence and energy were also relatively high: 1.652 and 1.397, respectively.

In more recent periods the structure of feature importance was different. For 2010s loudness was the most important (1.874), and then energy (0.484) and instrumentalness (0.424). In 2020s loudness was also dominant (1.544), which was followed by danceability (0.349), energy (0.294) and instrumentalness (0.235).

At the same time, the importance of acousticness and valence (which was high in the older periods) greatly decreased in 2020s. Negative values appeared for some features, e.g., danceability and speechiness in 2000s. It means the following: their random shuffling did not make the model perform worse and they were not useful predictors in the given model (*figure 7*).

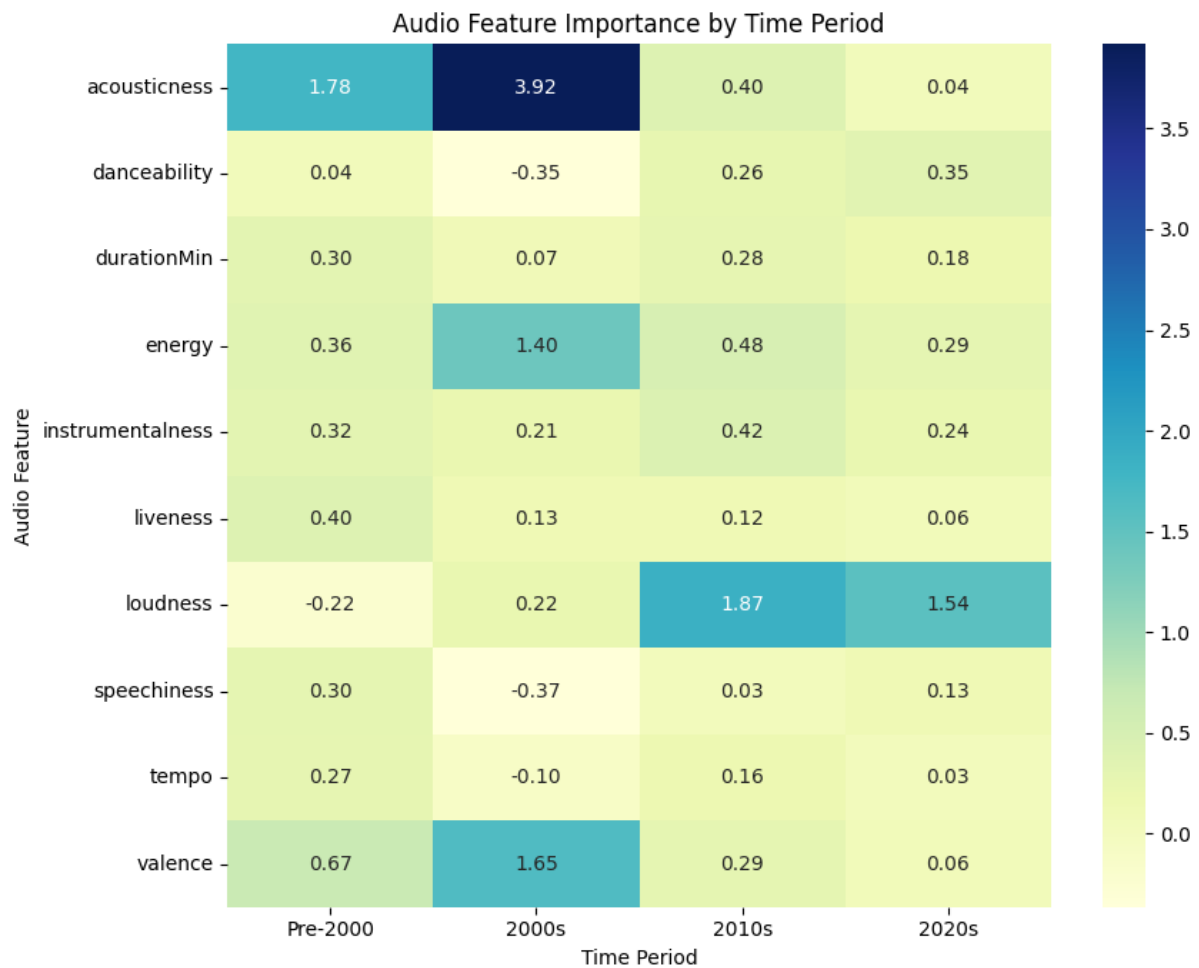


Figure 7 - Audio Feature Importance by Time Period

Discussion

Interpreting Predictive Performance

The predictive modelling results show that audio features alone have some kind of a predictive value, but their abilities are heavily limited. Linear Regression achieved $R^2 = 0.065$, whereas XGBoost improved on that score to ~ 0.121 . It means that the non-linear model was better at capturing more complex relationships between audio features and popularity but still explains just a little part of the variation of the target variable.

It is also worth noting that the hyperparameter optimisation practically did not improve performance. Optimised XGBoost achieved $R^2 = 0.12$, so an almost identical score compared to the base model. It suggests that the main limitation were not poorly chosen parameters, but the amount of information within audio features themselves. This is an important finding, because it suggests that increasing model complexity or adjusting the parameters cannot compensate for information that is not present in the predictor variables. In other words, the limitation appears to be connected more strongly to the available data than to the choice of the model itself.

This result is consistent with the previous EDA, which did not show any strong simple relationships between the majority of single audio features and popularity. Popularity of a song seem to depend on a larger number of factors, not only on the musical characteristics. Audio features can support predictions, but by themselves are not enough to build a model with high predictive accuracy. This also indicates that popularity should be understood as a multidimensional outcome. Two songs with relatively similar audio characteristics can still achieve very different levels of popularity because of differences in artist recognition, promotion, playlist exposure, release timing or other contextual factors that are not represented in this dataset.

Importance of Temporal and Genre Context

The highest improvement in predictive performance occurred after expanding the model by adding genre and releaseYear variables. Contextual XGBoost achieved $R^2 = 0.342$, whilst the optimised audio-only model got only 0.12. It means that the additional cultural and temporal context delivered information that the audio features did not contain.

The result suggests that the popularity of a song is not only connected to the audio features, but also to the release period and a larger genre context. Genre might indirectly represent differences in style, production and music reception. ReleaseYear, on the other hand, might reflect the trend changes and the characteristics of Spotify popularity score. The scale of the improvement is especially relevant here. The increase in R^2 from approximately 0.12 to 0.342 suggests that contextual information explains a substantial amount of variation that cannot be captured from the acoustic characteristics of a track alone. This strengthens the argument that popularity needs to be analysed within the environment in which a song was released rather than independently from it.

Although, it does not mean that genre and releaseYear directly measure the cultural aspects. In the context of this work, they're treated as contextual proxies. releaseYear, especially, requires special attention, because older tracks are significantly less represented in the dataset and might involve songs, that have been popular for a long time.

In practice, the results of the contextual model show that the predictions based on audio features only skips an important context part associated with the genre and time. Genre can also capture combinations of characteristics that may not be represented well by individual audio variables. For example, tracks belonging to different genres can share similar values of energy or danceability while still existing within very different musical conventions and audiences. Therefore, genre may provide the model with information about the broader musical context in which those audio characteristics should be interpreted.

How Popularity Determinants Changed Over Time

The importance analysis showed that the meaning of particular audio features was not stable across time periods. In pre-2000 and 2000s models acousticness played a bigger role. Valence and energy were also relatively high. In more recent years, the structure changed. For 2010s and 2020s loudness became the most important feature. Danceability also played a big role in 2020s.

This change suggests that the predictive relationship between the song characteristics and popularity depends on time. It does not mean that specific features cause popularity, but the model uses them to some extent based on a given temporal subset. One possible explanation is that the musical environment represented by each period is different. As the proportions of genres change, the predictive meaning of individual audio characteristics may also change. A feature such as danceability, for example, may become more informative in a period containing greater proportions of genres in which rhythmic characteristics are particularly prominent. Similarly, changes in loudness importance may partly reflect differences in production styles and in the types of genres represented in more recent periods.

The changing importance of instrumentalness can be interpreted in a similar way. The cultural analysis showed that more recent periods contain substantially more electronic, ambient and lo-fi tracks than the older periods. Because these genres can contain different levels and uses of instrumental content, instrumentalness may distinguish tracks differently depending on the genre composition of a particular period. Therefore, its changing importance does not necessarily mean that listeners became more or less interested in instrumental music. Instead, it may indicate that the relationship between instrumentalness and popularity changes as the wider composition of the music represented in the dataset changes.

Acousticness provides another example. Its greater importance in the older temporal models could indicate that this characteristic differentiated popularity more strongly within the musical environment represented by those periods. Its reduced importance in later models suggests that the same characteristic does not provide an equally useful signal across all periods. This supports the main argument of the temporal analysis: feature importance should not be interpreted as fixed or universal.

In practice, it means that one global model might hide differences that are only visible after diving the model into different periods. A global feature importance value averages relationships across tracks released under very different musical conditions. As a consequence, it may make a feature appear moderately important overall even when it is highly informative in one period and much less informative in another. The temporal models therefore provide additional interpretative value beyond the overall predictive model.

It is also worth noting that the predictive performance of the same temporal models was not the same. 2000s had the highest R^2 , whereas scores for 2010s and 2020s were lower.

At the same time, comparing these periods require caution because of different sample sizes. Pre-2000 and 2000s contain less observations, so their feature importance might be less stable. The particularly strong performance of the 2000s model should therefore not automatically be interpreted as evidence that popularity was inherently easier to predict during that decade. It may partly reflect the specific composition of the smaller sample or stronger relationships between particular characteristics within that subset. Differences in R^2 between periods can therefore reflect both genuine temporal differences and differences in the data available for each period.

Nevertheless, the general pattern supports the assumption that structural predictors of popularity change over time.

Cultural Interpretation

The results of cultural and temporal analysis show that changes in audio features occur alongside the changes in the genre structure of the dataset. Older periods were strongly dominated by rock, whereas there was more diversity in 2010s and 2020s (electronic, latin, hip-hop, ambient, lofi). At the same time, the average characteristics of songs also changed. In more recent years valence and durationMin decreased. On the other hand, danceability, speechiness and instrumentality increased.

These two findings are likely to be related to some extent. Changes in the genre composition of the dataset can change the average audio characteristics observed within a period. For example, the greater representation of hip-hop and related genres may contribute to higher speechiness, while the increased representation of electronic, ambient and lo-fi music may contribute to differences in instrumentality. Likewise, greater representation of rhythm-oriented genres may help explain the increasing relevance of danceability in the more recent temporal models. These relationships cannot establish that genre changes caused the observed audio trends, but they provide a contextual explanation for why the structure of the predictor's changes over time.

The reduction in average valence is also noteworthy. Rather than interpreting it as evidence that music or listeners became universally "less positive", it is more appropriate to understand it as a change in the characteristics of the tracks represented in the dataset. Similarly, the reduction in average track duration may reflect changing conventions in music production and distribution, but the available data does not make it possible to determine the precise cause of that change.

One should not interpret these results as direct measurements of cultural changes, though. The dataset does not contain variables describing social values or audience demographics. Genre and time period are only contextual proxies here. The results might be treated as an evidence of changes in the structure of music represented by the dataset and not as a full explanation of cultural changes in the society.

Therefore, the cultural interpretation of this project is deliberately limited to observable changes in musical structure and genre representation. The analysis shows that the musical context surrounding popularity is not constant, but it cannot determine whether those changes resulted from changing audience preferences, technological developments, industry practices, cultural trends or a combination of these factors.

Comparison with Previous Literature

The results of this project largely align with the previous studies presented in the Literature Review section. A limited accuracy of audio-only models corresponds to Interiano et al.'s (2018) results. In their work audio features only allowed for a prediction to some extent but did not give a high predictive performance.

Similarities can also be seen in relation to Saragih (2023). But direct comparisons of his values with the results of this study are not fully appropriate, because both studies use different datasets, models and the ways to define popularity. The important similarity is therefore not the exact numerical performance, but the broader conclusion that measurable audio characteristics provide useful predictive information without completely explaining song popularity.

The result of the contextual model is especially important. In Zangerle et al.'s (2019) work adding releaseYear alongside genre also brought a significant improvement of performance compared to an audio-only model. The improvement observed in this dissertation follows the same general pattern and provides further support for combining lower-level musical characteristics with higher-level contextual information.

Temporal results are aligned with the general direction shown by Petitbon and Hitchcock (2022). It supports the interpretation that both the genre composition and the meaning of defined audio characteristics might change with time. The present study extends this idea by showing that temporal differences are visible not only in genre composition and average musical characteristics, but also in the variables that predictive models rely on when estimating popularity.

Limitations

The most important limitation is the structure of the dataset. Great majority of the songs comes from 2010s and 2020s, and older periods are represented by a smaller number of observations. It means that the results of temporal models for pre-2000 and 2000s might be less stable and more prone to be affected by individual songs.

The second limitation is the possibility of survivorship bias. Older songs that are currently available on Spotify playlists might be the ones that have been popular for many years. Because of that the popularity of older songs should not be interpreted as an evidence that an average song released in that period of time was more popular. This bias may also affect the temporal feature importance results, because the older tracks contained in the dataset may represent an unusually successful subset of music from those periods rather than a representative sample of all songs released at the time.

Another problem is the lack of important external factors. The dataset does not contain any information on previous artist popularity, playlist placement, social media, marketing budget or actual number of streams. It might partially explain a limited performance of audio-only models. It also means that the contextual model should not be interpreted as containing the complete context of song popularity. Although genre and releaseYear considerably improve performance, many potentially influential variables remain unavailable.

Popularity is also a measurement specific for Spotify, so it cannot be treated as a universal measure of musical success. On top of that, genre in this case plays a role of a contextual proxy and is not a direct measure of culture.

Finally, all results are observational. Feature importance and model coefficients show predictive associations, not causal relations. Because of that one cannot state that a certain audio characteristic directly causes a decrease or an increase in popularity. This distinction is particularly important when interpreting the temporal results. A change in the importance of a feature such as instrumentality or danceability does not demonstrate that changing that characteristic would cause a song to become more or less popular. It only shows that the feature provided different amounts of predictive information within different temporal subsets.

Answering the Research Question

The results of this study allow to answer both research questions. In terms of RQ1, structural determinants of popularity change over time. Permutation importance shows that in older periods acousticness and valence were more important, whereas more recently it was loudness and danceability. The predictive structure changes with the time period. The results therefore suggest that the relationship between measurable musical characteristics and popularity should be understood as temporally dependent rather than fixed across the full dataset.

In terms of RQ2, Spotify audio features allow predictions of popularity, but to some limited extent. The best audio-only model, XGBoost, achieved R^2 of ~ 0.12 , which shows a weak predictive accuracy. After adding genre and releaseYear variables it increased to 0.342, showing that additional context improves the performance. This indicates that audio characteristics provide a useful starting point for prediction, while substantially stronger performance can be achieved when they are interpreted together with contextual information.

Final results support both assumptions of this dissertation: audio features have partial predictive value and their meaning changes over time. Taken together, these findings connect the two research questions: the ability to predict popularity depends not only on which variables are available, but also on the temporal and musical context in which those variables operate.

Conclusion

The goal of this research was to investigate to what extent Spotify audio features might be used in predicting the popularity of songs and whether the meaning of these features changes over time. To accomplish this, a full Data Analytics pipeline was implemented, including data cleaning, feature engineering, EDA, cultural and temporal analysis and predictive modelling with the use of Linear Regression and XGBoost.

The results show that audio features alone do have limited predictive value. Linear Regression achieved $R^2 = 0.065$ and XGBoost improved that score to ~ 0.121 . Hyperparameter optimisation did not improve the metrics. Much better results were achieved after expanding the model with genre and releaseYear variables, where R^2 increased to 0.342. It shows that contextual information gives the model significant value on top of the audio characteristics of the song.

The temporal analysis showed that the feature importance does not stay stable. Acousticness and valence were more prevalent in older periods, whereas in 2010s and 2020s loudness was more important.

Finally, the popularity of music cannot be fully explained just by its measurable audio characteristics. Although, the results show that these features have analytical value, and their predictive importance changes based on the time period and a broader analytical context.

Contribution

The main contribution of this dissertation is the combination of predictive modelling with the analysis of temporal and contextual changes into one coherent pipeline. The project is not limited to the evaluation of model performance only but uses feature importance for checking whether the meaning of individual audio features changes over time.

An additional value is extending the analysis of genre and releaseYear, which were treated as contextual proxies. It allowed showing that a model that includes a broader context achieves better results in terms of predictive performance compared to audio-only models.

As a result, this dissertation combines prediction, interpretability and temporal analysis, delivering a larger image of dependencies between the characteristics of music and its popularity.

Future Work

Future studies could, most importantly, use more balanced temporal dataset, which contains more songs from older periods. It would allow to limit the influence of uneven sample sizes and the survivorship bias, which make the feature importance between decades comparison harder.

Another extension would be adding external factors, which this dataset does not contain. It could be artist popularity, playlist placement, number of streams, social media activity, or information on promotions. Such variables could help in checking what part of popularity is connected to audio characteristics, and what part is connected to a broader social context.

In the future more Machine Learning models could also be implemented that utilise more detailed temporal division, but only if a larger number of observations for older periods is available. It would allow verification of whether the observed changes in predictive importance are the same in different models and datasets.

References

Interiano, M., Kazemi, K., Wang, L., Yang, J., Yu, Z. and Komarova, N.L. (2018). Musical trends and predictability of success in contemporary songs in and out of the top charts. Royal Society Open Science, [online] 5(5). <https://doi.org/10.1098/rsos.171274>

Harriman Samuel Saragih (2023). Predicting song popularity based on spotify's audio features: insights from the Indonesian streaming users. Journal of Management Analytics, 10(4), pp.693–709. <https://doi.org/10.1080/23270012.2023.2239824>

Petitbon, A.M. and Hitchcock, D.B. (2022). What Kind of Music Do You Like? A Statistical Analysis of Music Genre Popularity Over Time. Journal of Data Science, 20(2), pp.168–187. <https://doi.org/10.6339/22-jds1040>

Zangerle, E., Huber, R., Vötter, M. and Yang, Y.-H. (2019). HIT SONG PREDICTION: LEVERAGING LOW-AND HIGH-LEVEL AUDIO FEATURES. [online] <https://doi.org/10.5281/zenodo.3258042>

The above references have been generated with the aid of this tool:

<https://www.mybib.com/tools/harvard-referencing-generator>

Dataset used: <https://www.kaggle.com/datasets/solomonameh/spotify-music-dataset/data>

The code used in this submission was generated with the aid of AI. All the links to conversations with ChatGPT are present in the artefact provided.